

A SPECTRAL CHARACTERIZATION OF RESISTANCE-REGULAR GRAPHS*

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ABSTRACT. Let G be a connected simple graph and let R_G be its resistance distance matrix. Zhou, Wang and Bu introduced *resistance-regular* graphs, namely graphs for which all row sums of R_G are equal, and gave several equivalent conditions for this property. We present a self-contained spectral form of their characterization: a connected graph is resistance-regular if and only if the diagonal of the Moore-Penrose inverse L_G^+ of its Laplacian is constant. Equivalently, if $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$ and $q_2, \dots, q_n$ are orthonormal Laplacian eigenvectors, then

$$\sum_{k=2}^n \frac{q_k(v)^2}{\lambda_k}$$

is independent of the vertex v . This criterion gives a one-line recognition algorithm, an edge-local Foster-type condition, and, through Kirchhoff's edge formula for uniform spanning trees, a probabilistic interpretation: in a resistance-regular graph every vertex has the same expected degree in a uniform spanning tree. We also record structural consequences and examples. In particular, resistance-regular graphs on at least three vertices are 2-connected, all connected walk-regular graphs are resistance-regular, a complete multipartite graph is resistance-regular exactly when all parts have the same size, and regularity alone is not sufficient.

1. INTRODUCTION

Throughout the paper, graphs are finite, simple, undirected and connected unless stated otherwise. If every edge of a graph G is replaced by a unit resistor, the *resistance distance* r_{uv} between two vertices $u, v \in V(G)$ is the effective electrical resistance between them. The matrix

$$R_G = (r_{uv})_{u,v \in V(G)}$$

is the *resistance matrix* of G . Following Zhou, Wang and Bu [12], a graph is called *resistance-regular* if all row sums of R_G are equal. Equivalently, each vertex has the same total resistance distance to all other vertices.

The terminology is parallel to transmission-regularity for the ordinary distance matrix. Transmission-regular graphs include non-degree-regular examples, so it is natural to ask whether the electrical analogue behaves similarly. Zhou, Wang and Bu posed the problem of determining the resistance-regular graphs and, in particular, asked whether a nonregular resistance-regular graph exists [12]. Haritha and Chithra recently gave another eigenvector criterion and proved that resistance-regularity forces degree-regularity [8]. The present note extracts a compact spectral recognition theorem and organizes several consequences around it.

Let $L = L_G$ be the Laplacian of G and let $L^+ = L_G^+$ denote its Moore-Penrose inverse. The main criterion is

$$(1.1) \quad G \text{ is resistance-regular} \iff (L^+)_{vv} \text{ is independent of } v \in V(G).$$

Date: Manuscript draft, June 2026.

2020 Mathematics Subject Classification. 05C50, 05C12, 15A09, 94C15.

Key words and phrases. Resistance distance, resistance matrix, Laplacian matrix, Moore-Penrose inverse, Kirchhoff index, spanning trees.

*Supported by the Natural Science Foundation of Shandong Province (Grant No. ZR2024MA073).

This condition is not merely a restatement: it converts a condition on all pairwise effective resistances into a diagonal condition on one positive semidefinite matrix. It also gives the exact common value,

$$(L^+)_{vv} = \frac{\text{Kf}(G)}{n^2},$$

where $n = |V(G)|$ and $\text{Kf}(G) = \frac{1}{2} \sum_{u,v} r_{uv}$ is the Kirchhoff index.

The equivalence in (1.1), the ordinary inverse form, and the edge-local condition below are contained in the characterizations of Zhou, Wang and Bu [12, Theorem 12]. We present them here in a self-contained diagonal-and-eigenvector form, spell out the algorithmic and spanning-tree interpretations, and record several structural corollaries. Resistance-regularity is also equivalent to constant resistance curvature in the sense studied in [4, 11].

The results below should be read as a complete recognition theorem rather than as a finite forbidden-subgraph classification. The family is broad: it contains all connected walk-regular graphs, hence all connected vertex-transitive, distance-regular and strongly regular graphs. On the other hand, regularity by itself is not enough, as Example 6.5 shows.

2. PRELIMINARIES

Let $G = (V, E)$ have vertex set $V = \{1, \dots, n\}$. We write $u \sim v$ when $uv \in E$, and d_v denotes the degree of a vertex v . Its adjacency matrix is A , its diagonal degree matrix is D , and its Laplacian is

$$L = D - A.$$

The identity matrix is denoted by I , the all-one vector by $\mathbf{1}$, and $J = \mathbf{1}\mathbf{1}^\top$. Since G is connected,

$$(2.1) \quad \ker L = \text{span}\{\mathbf{1}\}, \quad LL^+ = L^+L = I - \frac{1}{n}J, \quad L^+\mathbf{1} = 0.$$

The resistance distance can be written in terms of L^+ as

$$(2.2) \quad r_{uv} = (L^+)_{uu} + (L^+)_{vv} - 2(L^+)_{uv}.$$

This formula is standard; it follows, for example, from the generalized-inverse formula for effective resistance [1, 2, 7, 9].

The Kirchhoff index is

$$\text{Kf}(G) = \frac{1}{2} \sum_{u,v \in V} r_{uv}.$$

If the Laplacian eigenvalues are

$$0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n,$$

then

$$(2.3) \quad \text{Kf}(G) = n \sum_{k=2}^n \frac{1}{\lambda_k} = n \text{tr}(L^+).$$

This spectral formula for the Kirchhoff index is classical [6].

We shall use the vector

$$g = \text{diag}(L^+) \in \mathbb{R}^n, \quad g_v = (L^+)_{vv}.$$

3. THE MAIN CHARACTERIZATION

Theorem 3.1 (Main criterion). *Let G be a connected graph on n vertices. The following statements are equivalent.*

- (1) G is resistance-regular.
- (2) $R_G \mathbf{1} = \rho \mathbf{1}$ for some scalar ρ .
- (3) The diagonal entries of L_G^+ are all equal.

(4) For every vertex v ,

$$(L_G^+)_{vv} = \frac{\text{Kf}(G)}{n^2}.$$

(5) If $q_2, \dots, q_n$ are orthonormal eigenvectors of L_G associated with $\lambda_2, \dots, \lambda_n$, then

$$\sum_{k=2}^n \frac{q_k(v)^2}{\lambda_k}$$

is independent of v .

When these conditions hold, the common row sum of R_G is

$$\rho = \frac{2 \text{Kf}(G)}{n}.$$

Proof. The equivalence of (1) and (2) is the definition. By (2.2) and $L^+ \mathbf{1} = 0$, the v -th row sum of R_G is

$$\begin{aligned} \sum_{u \in V} r_{vu} &= \sum_{u \in V} ((L^+)_{vv} + (L^+)_{uu} - 2(L^+)_{vu}) \\ &= n(L^+)_{vv} + \text{tr}(L^+). \end{aligned}$$

Thus the row sums of R_G are all equal if and only if the entries of $\text{diag}(L^+)$ are all equal. This proves (2) $\iff$ (3).

If the diagonal entries of L^+ are equal, then their common value is

$$\frac{\text{tr}(L^+)}{n} = \frac{\text{Kf}(G)}{n^2},$$

using (2.3). This proves (3) $\iff$ (4), and substituting this value into the row-sum formula gives

$$\rho = n \frac{\text{Kf}(G)}{n^2} + \frac{\text{Kf}(G)}{n} = \frac{2 \text{Kf}(G)}{n}.$$

Finally, choose an orthonormal Laplacian eigenbasis $q_1 = n^{-1/2} \mathbf{1}, q_2, \dots, q_n$. The spectral decomposition of the Moore-Penrose inverse is

$$L^+ = \sum_{k=2}^n \frac{1}{\lambda_k} q_k q_k^\top.$$

Taking diagonal entries gives

$$(L^+)_{vv} = \sum_{k=2}^n \frac{q_k(v)^2}{\lambda_k},$$

so (3) and (5) are equivalent. □

The matrix $L + n^{-1}J$ is nonsingular and satisfies

$$(3.1) \quad L^+ = \left(L + \frac{1}{n}J \right)^{-1} - \frac{1}{n}J.$$

Consequently the criterion can also be phrased using an ordinary inverse.

Corollary 3.2 (Ordinary inverse form). *Let*

$$X = \left(L_G + \frac{1}{n}J \right)^{-1}.$$

Then G is resistance-regular if and only if $X_{11} = \dots = X_{nn}$.

Remark 3.3. Theorems 3.1 and 3.2 are equivalent to the diagonal characterizations obtained in [12, Theorem 12]. The proof above emphasizes the direct row-sum identity

$$(3.2) \quad R_G \mathbf{1} = n \text{diag}(L_G^+) + \text{tr}(L_G^+) \mathbf{1},$$

which is often the shortest route to the criterion.

4. LOCAL AND PROBABILISTIC FORMULATIONS

The diagonal criterion has a useful local form involving only resistances across edges. This condition is also one of the equivalent formulations in [12, Theorem 12]; the proof below shows how it follows directly from the diagonal criterion.

Proposition 4.1 (Local Foster criterion). *For a connected graph G on n vertices, the following are equivalent.*

- (1) G is resistance-regular.
- (2) For every vertex v ,

$$(4.1) \quad \sum_{u \sim v} r_{uv} = 2 - \frac{2}{n}.$$

Proof. Let $g_v = (L^+)_{vv}$. From the diagonal entry of $LL^+ = I - n^{-1}J$ we have

$$(4.2) \quad d_v g_v - \sum_{u \sim v} (L^+)_{uv} = 1 - \frac{1}{n}.$$

If g is constant, then (2.2) and (4.2) give

$$\sum_{u \sim v} r_{uv} = 2 \left(d_v g_v - \sum_{u \sim v} (L^+)_{uv} \right) = 2 - \frac{2}{n}.$$

Thus resistance-regularity implies (4.1) by Theorem 3.1.

Conversely, suppose (4.1) holds. Expanding the left side with (2.2) gives

$$\sum_{u \sim v} r_{uv} = d_v g_v + \sum_{u \sim v} g_u - 2 \sum_{u \sim v} (L^+)_{uv}.$$

Using (4.2), the condition (4.1) is equivalent to

$$\sum_{u \sim v} (g_u - g_v) = 0,$$

that is, $(Lg)_v = 0$ for all v . Since G is connected, $\ker L = \text{span}\{\mathbf{1}\}$, hence g is constant. The result follows from Theorem 3.1. $\square$

Summing (4.1) over all vertices gives Foster's theorem [5]

$$(4.3) \quad \sum_{uv \in E(G)} r_{uv} = n - 1.$$

Conversely, Foster's theorem says that the average value of the left side of (4.1) is $2(n-1)/n$. Therefore resistance-regularity is exactly the condition that every vertex realizes this average.

There is also a spanning-tree interpretation. Let T be a uniformly random spanning tree of G . Kirchhoff's effective resistance formula for uniform spanning trees gives [10, Chapter 4]

$$(4.4) \quad \mathbb{P}(uv \in T) = r_{uv} \quad \text{for every edge } uv \in E(G).$$

Hence

$$\mathbb{E} \deg_T(v) = \sum_{u \sim v} r_{uv}.$$

Combining this with Theorem 4.1 gives the following equivalent answer.

Corollary 4.2 (Uniform spanning tree form). *A connected graph G on n vertices is resistance-regular if and only if*

$$\mathbb{E} \deg_T(v) = 2 - \frac{2}{n}$$

for every vertex v , where T is a uniformly random spanning tree of G .

5. RECOGNITION ALGORITHM

The preceding results give a direct polynomial-time recognition algorithm.

Proposition 5.1 (Recognition). *Given the adjacency matrix of a connected graph G on n vertices, resistance-regularity can be decided by one matrix inversion in $O(n^3)$ arithmetic operations.*

Proof. Form the Laplacian L and the matrix

$$M = L + \frac{1}{n}J.$$

The eigenvalues of M are $1, \lambda_2, \dots, \lambda_n$, so M is nonsingular. By [Theorem 3.2](#), G is resistance-regular if and only if the diagonal entries of M^{-1} are equal. The cost is dominated by the inversion of an $n \times n$ matrix. $\square$

For exact input, this test may be carried out over the rationals because $L + n^{-1}J$ has rational entries. Numerically, one can equivalently compute L^+ by an eigendecomposition and check whether the spread of $\text{diag}(L^+)$ is below a chosen tolerance.

6. EXAMPLES AND NON-EXAMPLES

6.1. Walk-regular graphs. A graph is *walk-regular* if, for every $k \geq 0$, the number of closed walks of length k starting at a vertex is independent of the vertex. Equivalently, the diagonal of A^k is constant for every $k \geq 0$.

Proposition 6.1. *Every connected walk-regular graph is resistance-regular.*

Proof. A connected walk-regular graph is regular; say $D = dI$. Since G is regular, J is a polynomial in A on a connected graph, and

$$L + \frac{1}{n}J = dI - A + \frac{1}{n}J$$

is also a polynomial in A after inversion on the finite spectrum. More explicitly, because $L + n^{-1}J$ is nonsingular, its inverse is a polynomial in $L + n^{-1}J$, hence a polynomial in A . The diagonal of every polynomial in A is constant by walk-regularity. By [Theorem 3.2](#), G is resistance-regular. $\square$

This recovers, in particular, the resistance-regularity of connected vertex-transitive graphs, distance-regular graphs, strongly regular graphs, cycles, complete graphs, hypercubes and many Cayley graphs. The walk-regular implication was already noted in [[12](#), Remark 2]; standard background on walk-regular and related spectral graph classes can be found in [[3](#)].

6.2. Complete multipartite graphs. The diagonal criterion gives a short classification of complete multipartite graphs.

Theorem 6.2. *Let $G = K_{n_1, \dots, n_m}$ be a complete multipartite graph, where $m \geq 2$, $n_i \geq 1$, and $n = n_1 + \dots + n_m$. Then G is resistance-regular if and only if*

$$n_1 = n_2 = \dots = n_m.$$

Proof. Let V_i be the part of size n_i . The Laplacian eigenstructure of $K_{n_1, \dots, n_m}$ is as follows. For each i , vectors supported on V_i with coordinate sum zero have eigenvalue $n - n_i$, and the subspace of vectors constant on each part and orthogonal to $\mathbf{1}$ has eigenvalue n .

Fix a vertex $v \in V_i$, and let e_v be the coordinate vector of v . Decomposing e_v into these orthogonal eigenspaces gives

$$(L^+)_{vv} = \frac{1 - 1/n_i}{n - n_i} + \frac{1/n_i - 1/n}{n}.$$

As a function of $t = n_i$, this equals

$$f(t) = \frac{t-1}{t(n-t)} + \frac{n-t}{tn^2} = \frac{n^2 - 2n + t}{n^2(n-t)}.$$

For $1 \leq t < n$,

$$f'(t) = \frac{n-1}{n(n-t)^2} > 0.$$

Thus the diagonal value $(L^+)_{vv}$ depends strictly increasingly on the size of the part containing v . By [Theorem 3.1](#), the graph is resistance-regular exactly when all part sizes are equal. $\square$

In particular, $K_{a,b}$ is resistance-regular if and only if $a = b$, and $K_{s,s,\dots,s}$ is resistance-regular for every $s \geq 1$.

6.3. Trees and cut vertices. Resistance-regular graphs have a strong connectivity property.

Proposition 6.3. *If G is resistance-regular and $n \geq 3$, then G has no cut vertex. Equivalently, every resistance-regular graph on at least three vertices is 2-connected.*

Proof. Suppose c is a cut vertex. Let H_1 be a smallest component of $G - c$, and let $h = |V(H_1)|$. Since $G - c$ has at least two components, $h \leq (n-1)/2$. Choose a vertex $u \in V(H_1)$ adjacent to c , and put $a = r_{uc} > 0$.

If $x \notin V(H_1) \cup \{c\}$, then every electrical connection from u to x passes through the cut vertex c , so resistance distances add:

$$r_{ux} = r_{uc} + r_{cx} = a + r_{cx}.$$

If $x \in V(H_1)$, the triangle inequality for the resistance metric [\[9\]](#) gives

$$r_{ux} - r_{cx} \geq -r_{uc} = -a.$$

Therefore, for the resistance row sums $S_y = \sum_x r_{yx}$,

$$\begin{aligned} S_u - S_c &= \sum_x (r_{ux} - r_{cx}) \\ &\geq -ha + (n-h-1)a + a \\ &= (n-2h)a > 0. \end{aligned}$$

Thus $S_u \neq S_c$, contradicting resistance-regularity. $\square$

Corollary 6.4. *A tree is resistance-regular if and only if it is K_1 or K_2 .*

Proof. The graphs K_1 and K_2 are resistance-regular. Every tree on at least three vertices has a cut vertex, so it is not resistance-regular by [Theorem 6.3](#). $\square$

6.4. Regularity is not sufficient.

Example 6.5. Let G be the cubic graph on vertices $0, 1, \dots, 7$ with edge set

$$\begin{aligned} E(G) = \{ &(0, 1), (0, 6), (0, 7), (1, 3), (1, 7), (2, 4), \\ &(2, 5), (2, 7), (3, 4), (3, 6), (4, 5), (5, 6) \}. \end{aligned}$$

Then G is 3-regular, but it is not resistance-regular. A direct calculation gives

$$\text{diag}(L^+) = \left(\frac{223}{672}, \frac{223}{672}, \frac{31}{96}, \frac{67}{224}, \frac{223}{672}, \frac{223}{672}, \frac{67}{224}, \frac{31}{96} \right),$$

which is not constant. By [Theorem 3.1](#), the graph is not resistance-regular.

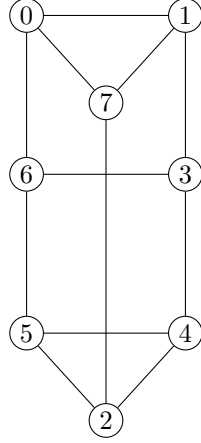


FIGURE 1. The cubic graph used in Example 6.5. The graph is degree-regular, but the diagonal entries of L^+ are not constant.

7. DISCUSSION

The criterion in Theorem 3.1 gives a concise answer to the question “which graphs are resistance-regular?”: exactly those connected graphs for which the diagonal of L_G^+ is constant. The equivalent local and probabilistic forms are often more intuitive:

$$\sum_{u \sim v} r_{uv} = 2 - \frac{2}{n} \iff \mathbb{E} \deg_T(v) = 2 - \frac{2}{n}.$$

Thus resistance-regularity says that each vertex has the average possible incident effective resistance, or equivalently the average possible expected degree in a uniform spanning tree.

Several structural classifications follow immediately in special graph classes. The family contains all connected walk-regular graphs but is strictly smaller than the class of regular graphs. Complete multipartite graphs are resistance-regular precisely in the balanced case, and the only resistance-regular trees are K_1 and K_2 . More generally, resistance-regular graphs on at least three vertices are 2-connected.

The main remaining direction is to translate the diagonal condition into more explicit structural classifications under natural hypotheses, for example for planar graphs, cographs, graph products, or graphs with few Laplacian eigenvalues. The diagonal criterion also suggests a computational search strategy: enumerate connected graphs in a desired class and test constancy of $\text{diag}(L_G^+)$ exactly over the rationals.

REFERENCES

- [1] R. B. Bapat, Resistance matrix of a weighted graph, *MATCH Commun. Math. Comput. Chem.* 50 (2004), 73–82.
- [2] R. B. Bapat, *Graphs and Matrices*, Universitext, Springer, London, 2010.
- [3] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Springer, New York, 2012.
- [4] K. Devriendt, A. Ottolini and S. Steinerberger, Graph curvature via resistance distance, *Discrete Appl. Math.* 348 (2024), 68–78.
- [5] R. M. Foster, The average impedance of an electrical network, in *Contributions to Applied Mechanics*, Edwards Brothers, Ann Arbor, 1949, pp. 333–340.
- [6] I. Gutman and B. Mohar, The quasi-Wiener and the Kirchhoff indices coincide, *J. Chem. Inf. Comput. Sci.* 36 (1996), 982–985.
- [7] I. Gutman and W. Xiao, Generalized inverse of the Laplacian matrix and some applications, *Bull. Acad. Serbe Sci. Arts Cl. Sci. Math. Natur.* 129 (2004), 15–23.
- [8] T. Haritha and A. V. Chithra, On the resistance regular graphs, arXiv:2405.18177v3, 2025.
- [9] D. J. Klein and M. Randić, Resistance distance, *J. Math. Chem.* 12 (1993), 81–95.
- [10] R. Lyons and Y. Peres, *Probability on Trees and Networks*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge, 2016.

- [11] W. Sun, Y. Yang and S.-J. Xu, On the minimum constant resistance curvature conjecture of graphs, *Discrete Appl. Math.* 389 (2026), 26–33.
- [12] J. Zhou, Z. Wang and C. Bu, On the resistance matrix of a graph, *Electron. J. Combin.* 23(1) (2016), Paper #P1.41, 10 pp.

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